

# Lab 2 — Agile Backlog & Sprint Simulation

Smart Lab Equipment & Slot Reservation Portal · PES University, Dept. of CSE · Problem Statement #01

Jira project **Smart Lab Portal** (key SLP), Scrum, company-managed. Backlog derived from the Lab 1 requirements table.

## 1. Epics and User Stories

Seven Epics covering the five functional requirements, both nonfunctional requirements and use case UC-13 from Lab 1. Twenty-four User Stories, 127 story points, estimated on the Fibonacci scale.

### Epic 1: Slot Reservation & Availability

Traces to FR-001

*Lets an authenticated student see live instrument availability and hold a slot of at most 2 hours, up to 7 days ahead, on an instrument whose calibration is valid.*

| Story                                  | User story                                                                                                                                            | Priority | Points | Sprint   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|
| <b>1.1</b> Real-Time Availability View | As a student, I want to see live availability for every instrument across the next 7 days, so that I can pick a free slot without walking to the lab. | High     | 5      | Sprint 1 |
| <b>1.2</b> Reserve a Slot              | As a student, I want to reserve a slot of up to 2 hours on a calibrated instrument, so that the instrument is guaranteed free when I arrive.          | High     | 8      | Sprint 1 |
| <b>1.3</b> Booking-Rule Enforcement    | As a lab administrator, I want to have requests over 120 minutes or more than 7 days ahead rejected, so that no one hoards scarce hardware.           | High     | 3      | Sprint 1 |
| <b>1.4</b> Reservation Token Issue     | As a student, I want to receive a unique reservation token on confirmation, so that I can prove my booking at the issue counter.                      | High     | 3      | Sprint 1 |

4 stories, 19 story points.

### Epic 2: Calibration Governance

Traces to FR-002

*Makes calibration state gate bookability rather than merely be displayed, so no reading is ever taken on an out-of-calibration instrument.*

| Story                                          | User story                                                                                                                                                                               | Priority | Points | Sprint   |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|
| <b>2.1</b> Record Calibration Event            | As a lab technician, I want to record date performed, next-due date and certificate ID against an instrument, so that every instrument's calibration state is on record.                 | High     | 5      | Sprint 1 |
| <b>2.2</b> Auto-Withdraw Expired Instruments   | As a faculty supervisor, I want to have an instrument leave the bookable inventory the moment its next-due date passes, so that no reading is taken on an out-of-calibration instrument. | High     | 8      | Sprint 1 |
| <b>2.3</b> Notify Affected Reservation Holders | As a student, I want to be told within 5 minutes if my future booking is voided by calibration expiry, so that I can rebook on another instrument in time.                               | Medium   | 5      | Backlog  |

3 stories, 18 story points.

### Epic 3: Equipment Custody

Traces to FR-003

*Ties physical custody of an instrument to the booking record through token validation and server-side check-out / check-in timestamps.*

| Story                                     | User story                                                                                                                                                         | Priority | Points | Sprint   |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|
| <b>3.1</b> Validate Token at Issue        | As a lab technician, I want to validate the student's reservation token before handing over an instrument, so that only the rightful booker takes it.              | High     | 5      | Sprint 2 |
| <b>3.2</b> Reject Invalid Tokens          | As a lab technician, I want to see expired, reused or wrong-instrument tokens rejected with a clear reason, so that I can refuse issue confidently at the counter. | High     | 3      | Sprint 2 |
| <b>3.3</b> Server-Side Custody Timestamps | As a auditor, I want to have check-out and check-in timestamps stored server-side against the reservation, so that a late return can be proven from the record.    | High     | 5      | Sprint 2 |

3 stories, 13 story points.

## Epic 4: Late-Return Penalties

Traces to FR-004

*Implements the disciplinary rule: overdue computation, a 30-minute grace window, strikes, and suspension on the third strike inside a rolling 30 days.*

| Story                                  | User story                                                                                                                                                          | Priority | Points | Sprint   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|
| <b>4.1</b> Compute Overdue at Check-In | As a lab technician, I want to have overdue duration computed automatically at check-in, so that I never have to judge lateness by hand.                            | High     | 5      | Sprint 2 |
| <b>4.2</b> Grace Window and Strike     | As a student, I want to have a 30-minute grace window before a strike is recorded, so that a queue at the issue counter does not penalise me.                       | High     | 5      | Sprint 2 |
| <b>4.3</b> Third-Strike Suspension     | As a faculty supervisor, I want to have booking privileges suspended for 14 days on a third strike in 30 days, so that repeat offenders stop blocking lab capacity. | High     | 8      | Sprint 2 |
| <b>4.4</b> Penalty Notifications       | As a student, I want to be notified within a minute of a strike or a suspension, so that I am never surprised by a blocked booking.                                 | Medium   | 3      | Backlog  |

4 stories, 21 story points.

## Epic 5: Cancellation & Waitlist

Traces to FR-005

*Recovers scarce lab capacity from abandoned bookings instead of leaving instruments idle.*

| Story                                | User story                                                                                                                                                             | Priority | Points | Sprint  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|---------|
| <b>5.1</b> Cancel a Reservation      | As a student, I want to cancel a reservation up to 2 hours before its start time, so that I am not penalised when my plans change.                                     | Medium   | 3      | Backlog |
| <b>5.2</b> Join a Waitlist           | As a student, I want to join the waitlist for a fully booked instrument-slot, so that I get a chance if someone cancels.                                               | Medium   | 5      | Backlog |
| <b>5.3</b> Sequential Waitlist Offer | As a waitlisted student, I want to be offered a released slot in waitlist order with 30 minutes to accept, so that the offer is fair and does not stall on one person. | Medium   | 8      | Backlog |

|                            |                                                                                                                                                |     |   |         |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---------|
| <b>5.4</b> No-Show Marking | As a faculty supervisor, I want to see unused, uncanceled reservations marked NO_SHOW, so that ghost bookings are visible in utilisation data. | Low | 3 | Backlog |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---------|

4 stories, 19 story points.

## Epic 6: Access Control, Audit & Concurrency

Traces to NFR-001, NFR-002

*Platform integrity: role-based access, an append-only evidence trail, and correct isolation on the contended reservation write.*

| Story                                   | User story                                                                                                                                                                                                         | Priority | Points | Sprint   |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|
| <b>6.1</b> Role-Based Access Control    | As a lab technician, I want to have calibration and penalty actions restricted to my role, so that students cannot alter their own records.                                                                        | High     | 5      | Sprint 1 |
| <b>6.2</b> Append-Only Audit Log        | As a auditor, I want to have every reservation, calibration, custody and penalty event logged immutably with actor, action, timestamp and before/after state, so that a suspension can be contested with evidence. | High     | 8      | Sprint 2 |
| <b>6.3</b> Twelve-Month Audit Export    | As a lab technician, I want to export 12 months of the audit log as CSV, so that I can answer an external audit without developer help.                                                                            | Low      | 3      | Backlog  |
| <b>6.4</b> Race-Free Concurrent Booking | As a student, I want to have exactly one of many simultaneous requests for the same slot succeed, within 200 ms, so that double-booking is impossible rather than merely unlikely.                                 | High     | 13     | Backlog  |

4 stories, 29 story points.

## Epic 7: Lab Utilisation Reporting

Traces to UC-13

*Gives the Faculty Supervisor visibility of how lab capacity is actually used.*

| Story                                   | User story                                                                                                                                       | Priority | Points | Sprint  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|---------|
| <b>7.1</b> Utilisation Report           | As a faculty supervisor, I want to see booked versus actually used hours per instrument, so that I can find where lab capacity is being wasted.  | Medium   | 5      | Backlog |
| <b>7.2</b> Filter Utilisation by Period | As a faculty supervisor, I want to filter utilisation by date range and department, so that I can justify new equipment purchases with evidence. | Low      | 3      | Backlog |

2 stories, 8 story points.

## 2. Sprint results

| Sprint            | Started            | Items | Points | Completed |
|-------------------|--------------------|-------|--------|-----------|
| SLP Sprint 1      | 02/Sep/26 12:11 PM | 7     | 37     | 7 of 7    |
| SLP Sprint 2      | 02/Sep/26 12:24 PM | 7     | 39     | 7 of 7    |
| Remaining backlog | —                  | 10    | 51     | —         |

**Sprint 1 goal.** A student can find and hold a slot on an instrument that is provably in calibration, and only a technician can change calibration data.

**Sprint 2 goal.** An instrument can only be issued against a valid token, and a late return produces a strike that is defensible from the audit log.

Both sprints were time-boxed to one week ending 09/Sep/26. Epics 1–4 closed completely, Epic 6 partially (13 of 29 points), Epics 5 and 7 untouched.

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Epic 1: Slot Reservation & Availability

Epic 2: Calibration Governance

Epic 3: Equipment Custody

Epic 4: Late-Return Penalties

Epic 5: Cancellation & Waitlist

Epic 6: Access Control, Audit & Concurrency

Epic 7: Lab Utilisation Reporting

Create epic

Plan your sprint

Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.

Create

0 work items | Estimate: 0

Backlog (10 work items)

Create a whiteboard to plan your work. Try

SLP-22 Story 2.3: Notify Affected Reservation Holders

SLP-23 Story 4.4: Penalty Notifications

SLP-24 Story 5.1: Cancel a Reservation

SLP-25 Story 5.2: Join a Waitlist

SLP-26 Story 5.3: Sequential Waitlist Offer

SLP-27 Story 5.4: No-Show Marking

SLP-28 Story 6.3: Twelve-Month Audit Export

SLP-29 Story 6.4: Race-Free Concurrent Booking

SLP-30 Story 7.1: Utilisation Report

SLP-31 Story 7.2: Filter Utilisation by Period

Epic 2: Calibration ... To Do 5

Epic 4: Late-Retur... To Do 3

Epic 5: Cancellatio... To Do 3

Epic 5: Cancellatio... To Do 5

Epic 5: Cancellatio... To Do 8

Epic 5: Cancellatio... To Do 3

Epic 6: Access Co... To Do 13

Epic 7: Lab Utilisat... To Do 5

Epic 7: Lab Utilisat... To Do 3

Create

### 3. Burndown charts

In both charts the Remaining Values line falls from the committed total to zero within minutes of the sprint starting and then runs flat along the zero axis for the rest of the one-week time box, while the guideline descends slowly to meet it on 09/Sep. The line sits below the guideline not because the team was fast but because seven days of planned work were transacted in about ten minutes of elapsed time, so neither chart supports a velocity claim. Read as a real sprint, the shape is the signature of severe under-commitment.

#### SLP Sprint 1 – 37 story points



SLP Sprint 2 — 39 story points

